Salary Prediction

PGP DSBA - Capstone Project Report

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1. Introduction

1.1 Description: - Salary Prediction of Delta Ltd.: -

Business Context

Business Problem: To ensure there is no discrimination between employees, it is imperative for the Human Resources department of Delta Ltd. to maintain a salary range for each employee with similar profiles Apart from the existing salary, there is a considerable number of factors regarding an employee's experience and other abilities to which they get evaluated in interviews. Given the data related to individuals who applied in Delta Ltd, models can be built that can automatically determine salary which should be offered if the prospective candidate is selected in the company. This model seeks to minimize human judgment with regard to salary to be offered.

Object

The objective of this exercise is to build a model, using historical data that will determine an employee's salary to be offered, such that manual judgments on selection are minimized. It is intended to have a robust approach and eliminate any discrimination in salary among similar employee profiles

1.2 Data Description

The data contains the different attributes of employee Salary Prediction to hire for Delta Ltd. The detailed data dictionary is given below.

Target variable: Expected_CTC

1.3 Data Dictionary

IDX	Index
Applicant_ID	Application ID
Total_Experience	Total industry experience
Total_Experience_in_field_applied	Total experience in the field applied for (past work experience that is relevant to the job)
Department	Department name of current company
Role	Role in the current company
Industry	Industry name of current field
Organization	Organization name
Designation	Designation in current company
Education	Education
Graduation_Specialization	Specialization subject in graduation
University_Grad	University or college in Graduation
Passing_Year_Of_Graduation	Year of passing Graduation
PG_Specialization	Specialization subject in Post-Graduation

University_PG	University or college in Post-Graduation
Passing_Year_Of_PG	Year of passing Post Graduation
PHD_Specialization	Specialization subject in Post-Graduation
University_PHD	University or college in Post Doctorate
Passing_Year_Of_PHD	Year of passing PHD
Curent_Location	Curent Location
Preferred_location	Preferred location to work in the company applied
Current_CTC	Current CTC
Inhand_Offer	Holding any offer in hand (Y: Yes, N:No)
Last_Appraisal_Rating	Last Appraisal Rating in current company
No_Of_Companies_worked	No. of companies worked till date
Number_of_Publications	Number of papers published
Certifications	Number of relevant certifications completed
International_degree_any	Hold any international degree (1: Yes, 0: No)
Expected_CTC	Expected CTC (Final CTC offered by Delta Ltd.)

Table 1 Data Dictionary

2. Data Report

2.1 Data Overview (rows, columns, descriptive details).

- The dataset contains information 29 attributes and 25000 rows/entries.
- Data type of the columns Numerical (int64 & floa64) and Categorical (object)
- Int64: IDX,Application_ID,Total_Experience, Current_CTC, No.of companies worked, No.of Publications, Certifications and International_degree_any.
- float64: Passing_Year of Graduation/PG/PHD.
- object: Department, Role, Organization, Designation, Inhand_Offer, Last Appraisal Rating.
- Data in each row corresponds to the Expected salary of an individuals applied for an interview of a company Delta.Ltd.
- Apart from the existing salary all the remaining details (experience, education, salary location) were collected from the interviews.

2.2 Statistical summary of the data: -

	count	mean	std	min	25%	50%	75%	max
Total_Experience	25000.0	1.249308e+01	7.471398e+00	0.0	6.000000e+00	1.200000e+01	1.900000e+01	25.0
Total_Experience_in_field_applied	25000.0	6.258200e+00	5.819513e+00	0.0	1.000000e+00	5.000000e+00	1.000000e+01	25.0
Current_CTC	25000.0	1.760945e+06	9.202125e+05	0.0	1.027312e+06	1.802568e+06	2.443883e+06	3999693.0
No_Of_Companies_worked	25000.0	3.482040e+00	1.690335e+00	0.0	2.000000e+00	3.000000e+00	5.000000e+00	6.0
Number_of_Publications	25000.0	4.089040e+00	2.606612e+00	0.0	2.000000e+00	4.000000e+00	6.000000e+00	8.0
Certifications	25000.0	7.736800e-01	1.199449e+00	0.0	0.000000e+00	0.000000e+00	1.000000e+00	5.0
International_degree_any	25000.0	8.172000e-02	2.739431e-01	0.0	0.000000e+00	0.000000e+00	0.000000e+00	1.0
Expected_CTC	25000.0	2.250155e+06	1.160480e+06	203744.0	1.306278e+06	2.252136e+06	3.051354e+06	5599570.0
Percentage of Hike	25000.0	2.512749e+01	8.803702e+00	0.0	1.959996e+01	2.499998e+01	2.999999e+01	45.6

Table 2 Statistical Summary of the data set

- It is a statistical summary of the quantitative analysis.
- Average of total experience is 12.5 years approximately and the range is 0 – 25 (min – max) but the relevant field experience is 6.26 years.
- Average current_ctc is ₹176,095 and maximum ₹3.99 million.
- The average expected salary is ₹225,015 and maximum salary is ₹5.6 million. Usually, some candidates having high compensation.

2.3 Understanding of attributes (variable info, renaming if required)

- we understood the data provided is relating to Applicant details, Experience, Education and Salary.
- Renamed/Corrected these columns Current_Location,PG_Specialization, Phd_Specialization and University_Graduation.

3. Exploratory Data Analysis

3.1 Univariate analysis (distribution and spread for every continuous attribute, distribution of data in categories for categorical ones)

- Total_Experience data is balanced across various levels of experience.
- Most of the candidate's experience is irrelevant from 5 to 25 and only few individuals have relevant between 0 -5.
- Current_CTC is not normal distribution and right-skewed. Only few individuals receiving the better CTC currently so need to concentrate on them only.
- Many of the candidates worked in 2 or 3 companies and very less count for those who has zero work experience (i.e Freshers).
- It has fair and even counts across 0-8 publications.
- Majority of candidates have 0 certifications and some of them has 1-5 certifications. Candidates are recommendable to upskill their knowledge.
- No applicants have the international degree but only few individuals had 1. So, it is highly imbalanced.

- It is right skewed and distribution is not a normal. Only fewer candidates are expecting the higher compensation.

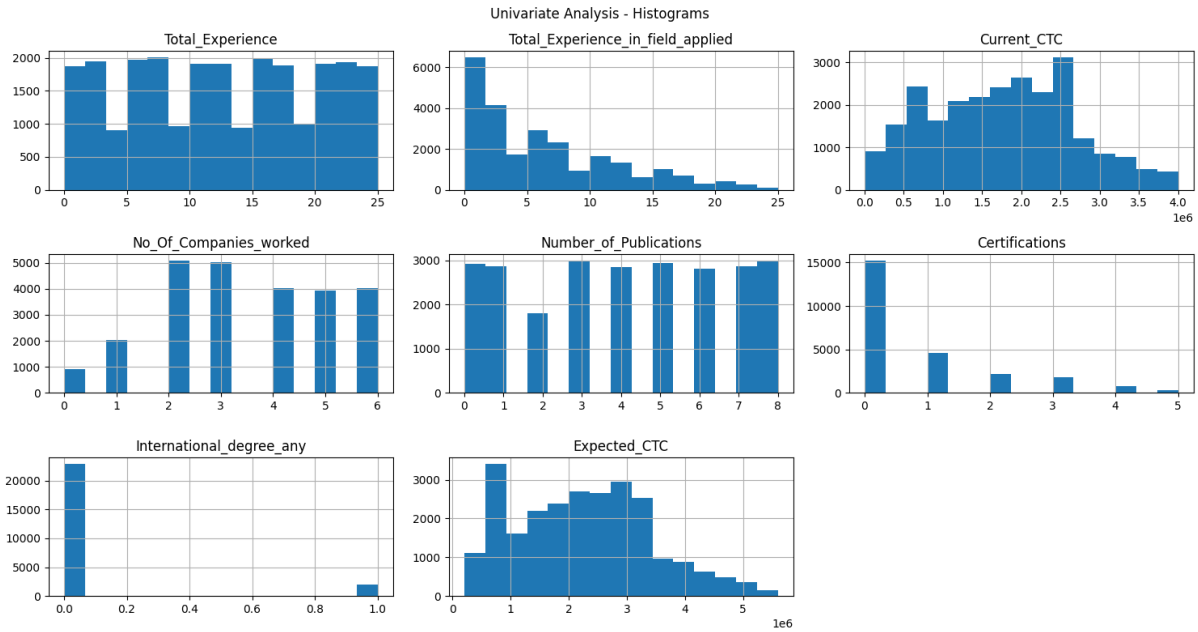

Figure 1 Univariate Analysis

3.2 Bivariate analysis (relationship between different numerical variables, correlations).

Last_Apprailsal_Rating: -

- Rating B was found in many profiles.
- Candidates with A, B, C ratings are more comparatively.
- We have 4191 candidates as Key_Performer last appraisal rating so need to concentrate on these profiles because they are hard workers and might have relevant field.

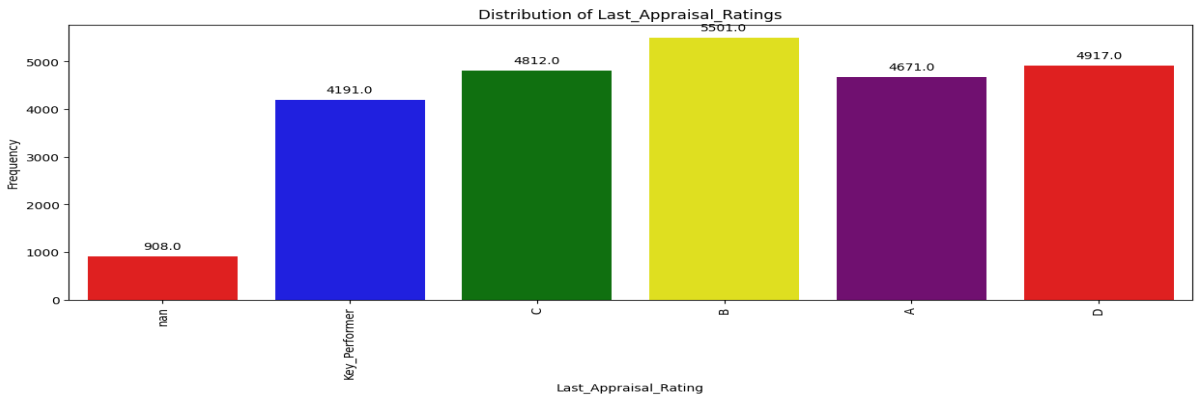

Figure 2 Last Appraisal Rating

Current_CTC: -

- It is not a normal distribution and skewed to the right-hand side.

- The highest frequency of the values lies in between 0.5M and 2.6M so they are receiving better Current_CTC. Therefore, it needs to concentrate on these individuals current_ctc.

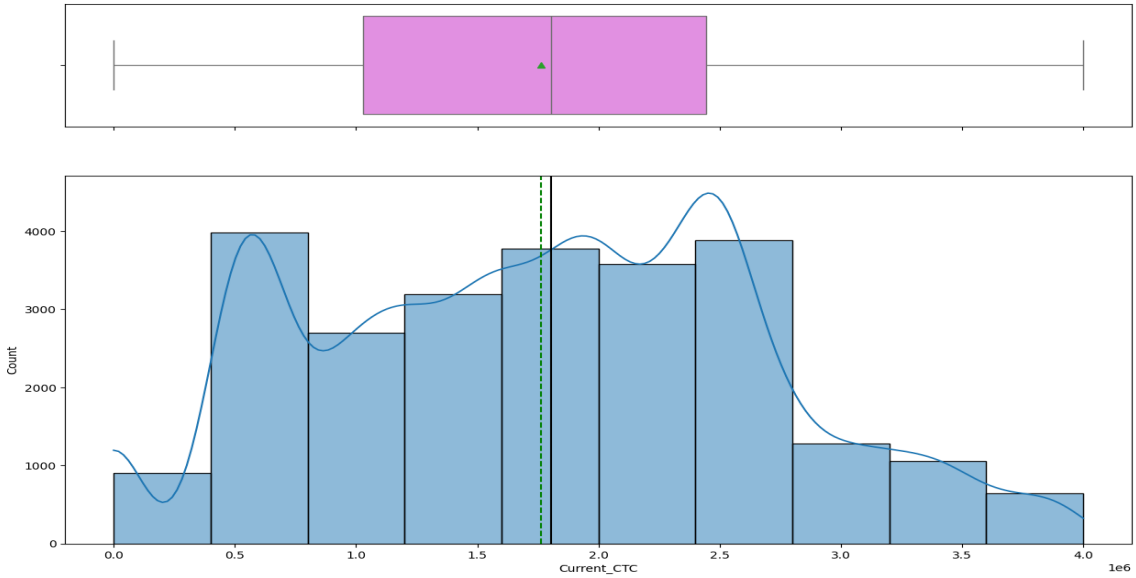

Figure 3 Current_CTC

3.3 Bivariate/Multivariate analysis (relationship between different numerical variables, correlations).

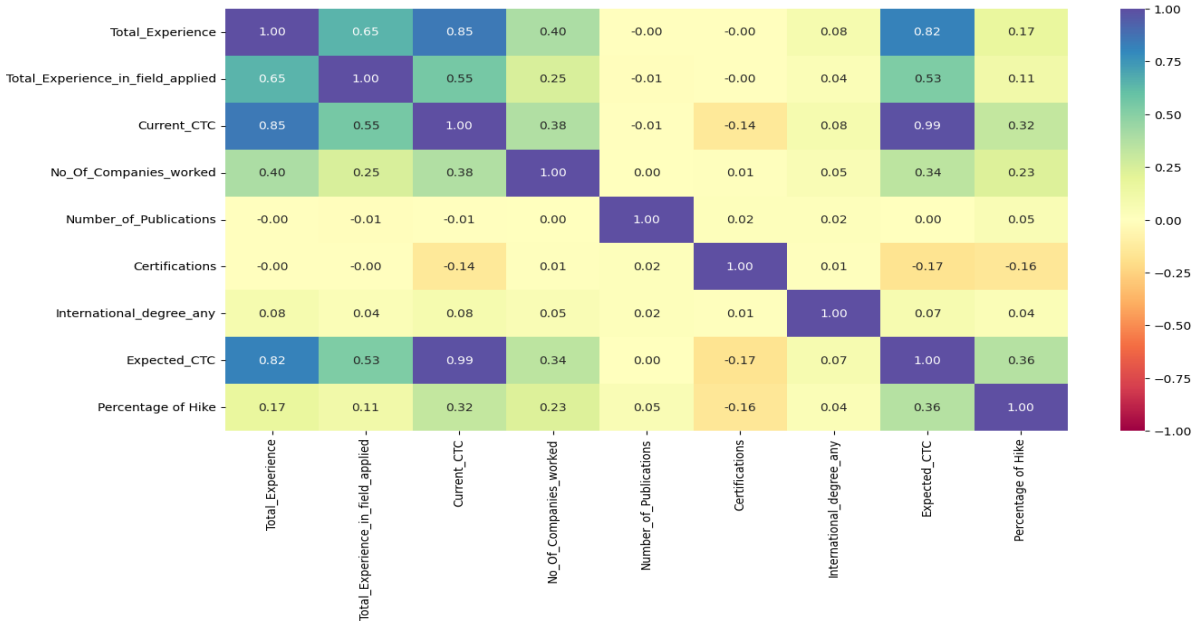

Figure 4 Heatmap

- Current_CTC and Expected_CTC variables are having higher correlation at 0.99. Hence, candidates will expect the high compensation based on their current_ctc.
- Total_Experience and Current_CTC attributes are more correlated each other at 0.85. So, more experience generally having higher present compensation.

- At 0.82 Total_Experience and Expected_CTC logically high experienced individuals will look into high compensation.
- Total_Experience and Total_Experience_in_field_applied have relation together at 0.65. Mostly, Total_Experience includes the relevant experience in field applied.
- No. of publications and Certifications shows no correlation (i.e 0.00) and International_degree_any have very low correlation (i.e 0.1). Meaningfully, these parameters have no impact on compensation as per the data set provided.
- No. of publications, Certifications, Percentage of Hike and Expected_CTC variables are negatively correlated. We will find them in less experienced profiles.

4. Data Cleaning and Pre-processing

4.1 Removal of unwanted variables

As per the dataset provided, the variables IDX & Applicant_ID have no impact on the output so it has dropped.

4.2 Missing Value treatment

PHD_Specialization, University_PHD, Passing_Year_of_PHD, PG_Specialization, University_PG and Passing_Year_of_PG had missing over 30% so it has replaced by Others as per the column requirement in dataset.

4.4 Outlier treatment

The below image helps to execute the outlier's treatment.

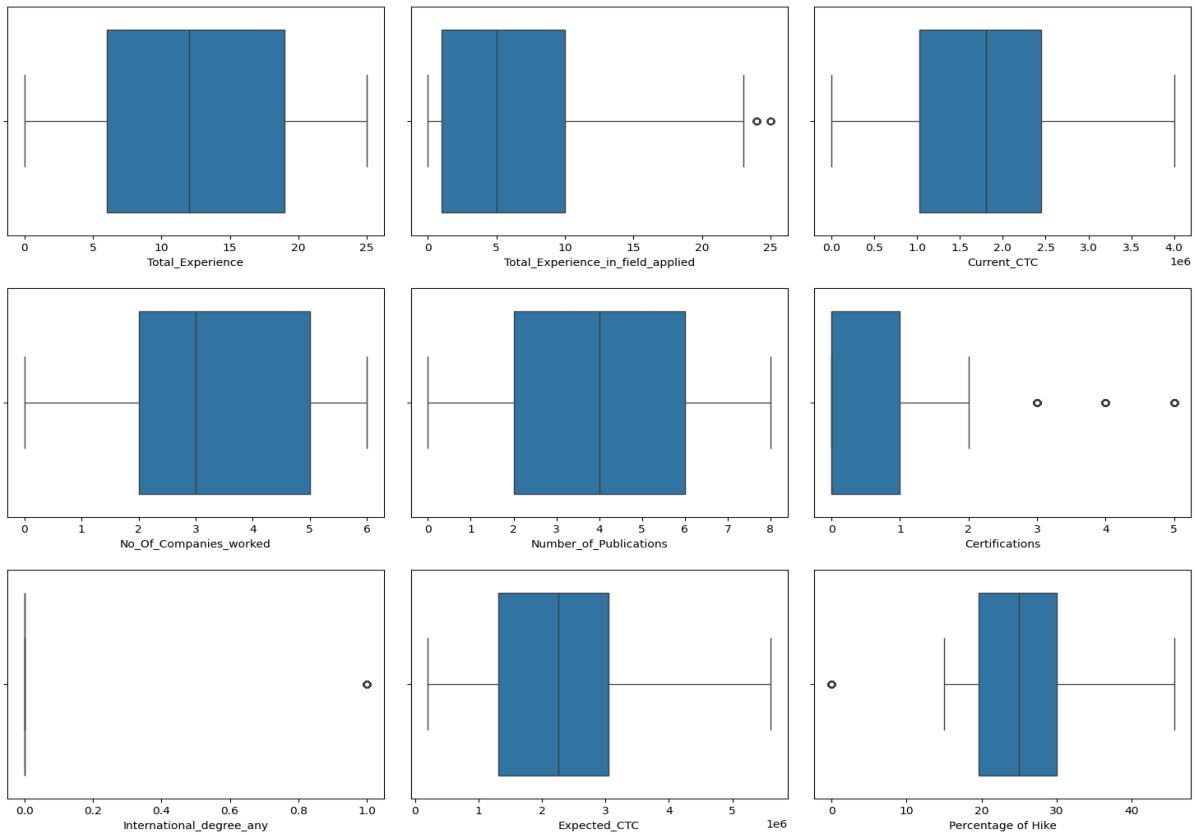

Figure 5 Bivariate Analysis

- No outliers for these variables are boxplots of Total_Experience, Current_CTC, No_of_Companies_worked, No_of_Publications and Percentage of Hike. So, no action needed.
- Total_Experience_in_field_applied has two outliers which are above 23 years.
- Certifications show 3 outliers which means the candidates are having 3-5 certificates. However, these are useful only when it is relevant to the experience.
- Few applicants did international degree any so it has only one outlier.
- These candidates are freshers so Percentage of Hike attribute reflecting an outlier as "0".

4.5 Variable transformation

Current_CTC

- Log transformed of Current_CTC has applied to reduce the right-skewness. Mean and Median lines are close together means the outliers reduced.
- It has only one outlier which is near by zero or zero so there is no major impact.

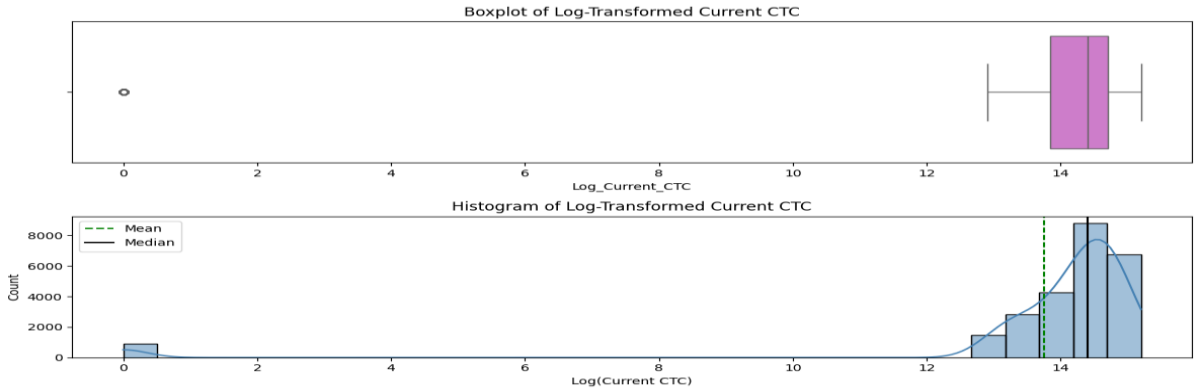

Figure 6 Log Transformation Current_CTC

Expected_CTC

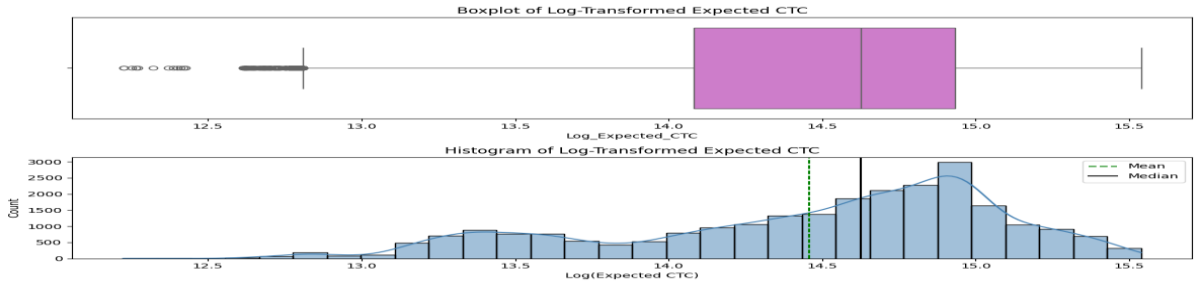

Figure 7 Log Transformation Expected_CTC

- Expected_CTC indicated many outliers so applied log transformation for Variable is successful.
- It is almost close to the normal distribution and left skewed.

4.6 Addition of new variables

Added few variables based on the requirement.

- Added Percentage of Hike column to calculate the Hike ratio of every individual.
- A new variable (Total_Experience_in_field_applied ratio) included to calculate whether the total experience is relevant or not.

4.7 Clustering Technique

Yes, we have outliers in using the clusters and adding the screen shots.

The Current_CTC and Expected_CTC variables consists of more outliers

- Many Candidates with one year experience had compensation over the range. So, need to give a chance to apply whoever within the range.
- Also, the candidates with one year experience are expecting the more compensation which is beyond the actual compensation offered to the designation.

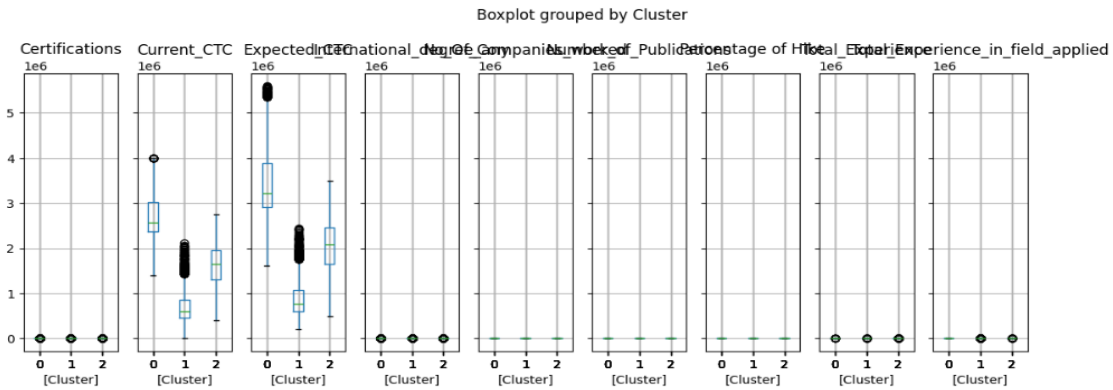

Figure 8 Cluster boxplot

- The Delta Ltd. is offering the less compensation to the applicants with internation degree any so it is better to increase the count. Indeed, these candidates completed global studies so it is helpful in case of directly dealing with clients or to enhance the Global market.
- People under the lesser than last appraisal rating are freshers or lower performers and above the last appraisal rating are expecting the higher compensation so it is cost to the company. Which is very critical to focus on it in long run.

5. Model Building: -

5.1 K - Means: - Elbow Method

This is an Elbow Method plot used to determine the optimal number of clusters (k) for K-Means clustering, based on the Expected_CTC (likely a cost-to-company or salary-related metric).

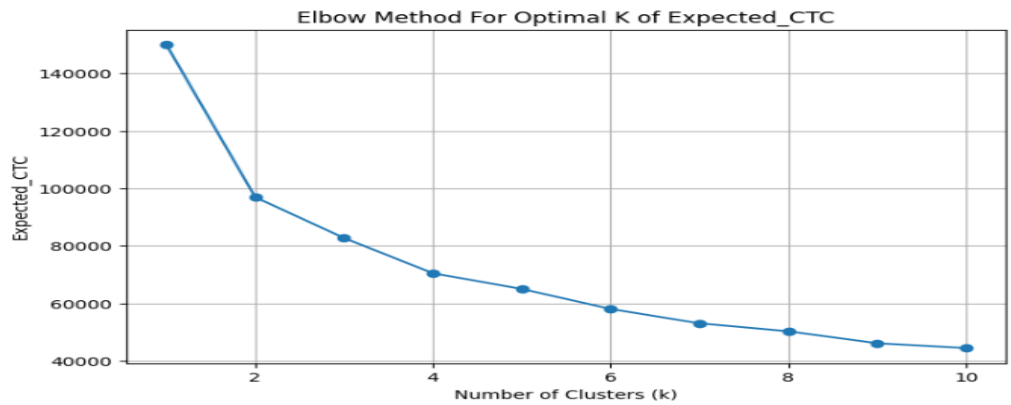

Figure 9 Elbow

- Though the bend is not coming out clearly as there are many bends, let us look at 1 cluster and 3 clusters. The Optimal clusters (K=4) are 4.
- These points balance model simplicity and performance. After k = 3, the decrease becomes more gradual and less significant.

5.2 Linear Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           Expected_CTC   R-squared:                       0.997
Model:                            OLS   Adj. R-squared:                  0.997
Method:                 Least Squares   F-statistic:                 1.913e+04
Date:                Sun, 18 May 2025   Prob (F-statistic):               0.00
Time:                        07:05:23   Log-Likelihood:            -2.1908e+05
No. Observations:               17500   AIC:                         4.387e+05
Df Residuals:                   17225   BIC:                         4.408e+05
Df Model:                         274
Covariance Type:            nonrobust
==============================================================================================
                                         coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                               1.081e+05   1.52e+04      7.090      0.000    7.82e+04    1.38e+05
Total_Experience                   -4678.1239    203.116    -23.032      0.000   -5076.252   -4279.996
Total_Experience_in_field_applied     46.0549    113.931      0.404      0.686    -177.261     269.371

No Experience_Fresher               1.475e+05   3614.793     40.796      0.000     1.4e+05    1.55e+05
==============================================================================
Omnibus:                     6660.699   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            88210.976
Skew:                           1.457   Prob(JB):                         0.00
Kurtosis:                      13.606   Cond. No.                     7.59e+22
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.2e-29. This might indicate that there are
strong multicollinearity problems or that the design_matrix is singular.
```

Figure 10 OLS Regression Results

5.2.1 Model Performance Check

It has two types of performance to verify.

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	66159.155912	46439.056764	0.996725	0.996669	3.578321

Table 3 model performance train set

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	66400.151442	46683.134184	0.996783	0.996653	3.556775

Table 4 model performance test set

5.2.2 Interpretation: -

- MAPE has excellent Predictive accuracy or error < 5%.
- R^2 values near to 1 on both training and test sets so the model explains almost all the variability. Therefore, it suggests the model is well-suited for the data and no underfitting.
- RMSE and MAE values are very close across training and test sets so it has low variance/generalization error.
- MAPE < 4% so it has strong real-world predictive performance, especially in cost/CTC-related regression tasks.
- The model is performing exceptionally well, with both high accuracy and generalizability.
- Adjusted R-squared ≈ R-squared. It Implies most predictors contribute meaningfully to the model so no unnecessary features.

Variance_Inflation_Factor: - These attributes are called inflation factors (291) which has been increasing the Expected _CTC.

	feature	VIF
0	const	913.993779
1	Total_Experience	9.071146
2	Total_Experience_in_field_applied	1.754577
3	Current_CTC	7.161035
4	No_Of_Companies_worked	1.412624
...	...	...
286	Last_Appraisal_Rating_C	9.023270
287	Last_Appraisal_Rating_D	9.153316
288	Last_Appraisal_Rating_Key_Performer	1.953925
289	Last_Appraisal_Rating_nan	inf
290	No Experience_Fresher	inf

291 rows × 2 columns

Table 5 VIF values

- Removed un-necessary columns and run the OLS (i.e OLS – Ordinary Least Square Regression) regression until all the p-values become under 5.
- Since the P Value is < 0.05 so all the variables randomly effecting the Expected salary of the candidates.

5.2.3 Final Model Summary: -

```
                            OLS Regression Results
==============================================================================
Dep. Variable:            Expected_CTC   R-squared:                       0.986
Model:                             OLS   Adj. R-squared:                  0.985
Method:                  Least Squares   F-statistic:                 1.186e+04
Date:                 Sun, 18 May 2025   Prob (F-statistic):               0.00
Time:                         08:22:20   Log-Likelihood:            -2.3207e+05
No. Observations:                17500   AIC:                         4.643e+05
Df Residuals:                    17399   BIC:                         4.651e+05
Df Model:                          100
Covariance Type:             nonrobust
==========================================================================================
                                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                           1.398e+05   7476.496     18.694      0.000    1.25e+05    1.54e+05
Total_Experience               -9819.0619    348.591    -28.168      0.000   -1.05e+04   -9135.789
Current_CTC                         1.3239      0.003    496.018      0.000       1.319       1.329
No_Of_Companies_worked         -7816.3304    722.136    -10.824      0.000   -9231.790   -6400.871
Certifications                 -1.342e+04   1046.206    -12.831      0.000   -1.55e+04   -1.14e+04
Department Analytics/BI        -2.047e+04   4741.646     -4.316      0.000   -2.98e+04   -1.12e+04
Inhand_Offer_Y                  5.057e+04   2785.252     18.155      0.000    4.51e+04     5.6e+04
Last_Appraisal_Rating_B         9.528e+04   2668.662     35.703      0.000       9e+04    1.01e+05
Last_Appraisal_Rating_Key_Performer 1.504e+05   3343.706     44.975      0.000    1.44e+05    1.57e+05
==============================================================================
Omnibus:                     3202.271   Durbin-Watson:                   2.006
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            13054.069
```

Table 6 OLR regression Final

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	139023.817921	104714.62595	0.985537	0.985446	6.211101

Table 7 final OLS Train set

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	137582.539182	104421.594522	0.98619	0.985986	6.162457

Table 8 final OLS Test set

5.3 Decision Tree Model

Decision tree model for prediction or further analysis of the Salary predictions.

```
DecisionTreeRegressor
DecisionTreeRegressor(min_impurity_decrease=0.01, min_samples_leaf=5,
                      random_state=1)
```

Table 9 decision tree model

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.0	0.0	1.0	1.0	0.0

Table 10 train set

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.40292	0.358622	-1.123168	-1.881443	93.307633

Table 11 test set

- The Decision tree model with default parameters is overfitting the train data.
- Let's see if it can reduce overfitting and improve performance on test data by tuning hyperparameters.

5.4 Random Forest Model

Random Forest Model for prediction or further analysis of the Salary predictions.

```
RandomForestRegressor
RandomForestRegressor(random_state=1)
```

Table 12 Random model

Checking model performance on training set.

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.108714	0.088341	0.866081	0.857032	70.240053

Table 13 train data

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.388639	0.344362	-0.975329	-1.680803	99.315509

Table 14 test data

- Training set performance is good due to lower error (MAPE) and higher R-Square for prediction.

5.5 Gradient Boosting Regressor Model

This is Gradient Boosting Regressor Model for prediction or further analysis of the Salary predictions.

```
GradientBoostingRegressor
GradientBoostingRegressor(random_state=1)
```

Table 15 Gradient Boosting Regressor Model

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.040768	0.032881	0.981167	0.979895	24.186933

Table 16 GB Model performance on Training set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.384474	0.340478	-0.933223	-1.62366	92.225739

Table 17 GB Model performance on Test set

Also, R2 is negative in GB Regressor Model. It can't be selected as best model.

5.6 XGBoost Regressor Model

XGBoost Regressor Model is the best model, as it has the least MAPE, based on the testing set.

```
XGBRegressor
XGB Saved to this PC se_score=None, booster=None, callbacks=None,
colsample_bylevel=None, colsample_bynode=None,
colsample_bytree=None, device=None, early_stopping_rounds=None,
enable_categorical=False, eval_metric=None, feature_types=None,
gamma=None, grow_policy=None, importance_type=None,
interaction_constraints=None, learning_rate=None, max_bin=None,
max_cat_threshold=None, max_cat_to_onehot=None,
max_delta_step=None, max_depth=None, max_leaves=None,
min_child_weight=None, missing=nan, monotone_constraints=None,
multi_strategy=None, n_estimators=None, n_jobs=None,
num_parallel_tree=None, random_state=1, ...)
```

Table 18 XGB Regressor Model

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.00095	0.000672	0.99999	0.999989	0.558363

Table 19 XGBoost Performance on training set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.353711	0.305792	-0.636231	-1.2206	78.067886

Table 20 XGBoost Performance on test set

5.2 Test your predictive model against the test set using various appropriate performance metrics.

Expected_CTC has classified in to two based on the median value. If the salary is above the median value, it is called High_CTC otherwise Low_CTC. Below is the table.

High_CTC	count
0	12500
1	12500

Table 21 High_CTC classification

This is the accuracy of the expected salary for prediction.

	precision	recall	f1-score	support
0	1.00	0.99	0.99	2507
1	0.99	1.00	0.99	2493
accuracy			0.99	5000
macro avg	0.99	0.99	0.99	5000
weighted avg	0.99	0.99	0.99	5000

Table 22 Accuracy by expected ctc classification

- All metrics are extremely high (≥ 0.99), indicating the model is nearly perfect at distinguishing between class 0 and class 1. So, it is performed well.
- Support values (2507 and 2493) are nearly equal, so accuracy is a reliable indicator to balance it.
- F1-scores of 0.99 mean that the model maintains a great balance between precision and recall.

5.3 TEST FOR LINEARITY AND INDEPENDENCE: -

	Actual Values	Fitted Values	Residuals
4289	3109048	3.174310e+06	-65261.711805
19621	2067059	2.059075e+06	7984.280082
14965	4915655	4.816938e+06	98716.869731
12321	1959755	1.962288e+06	-2532.785735
6269	4514894	4.206434e+06	308459.682418

Table 23 actual vs residual values

let's plot the fitted values vs residuals

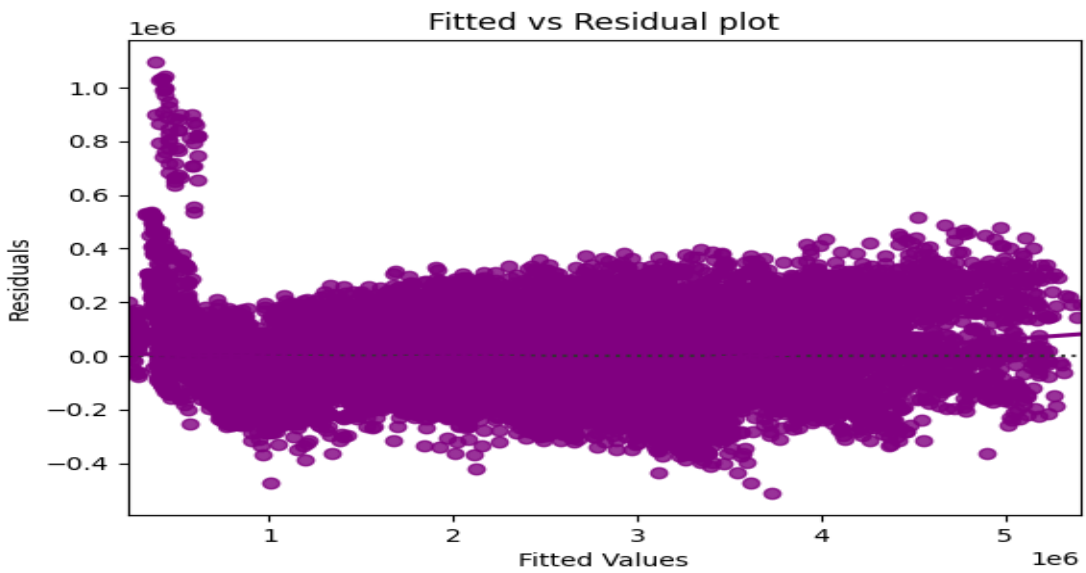

Figure 11 Fitted vs Residual plot

The above values are determined from the linear regression. It is the comparison of actual, fitted and residual values. It helps to minimize the variance while prediction.

5.4 TEST FOR NORMALITY: -

- We will test for normality by checking the distribution of residuals, by checking the Q-Q plot of residuals, and by using the Shapiro-Wilk test.
- If the residuals follow a normal distribution, they will make a straight line plot, otherwise not.
- If the p-value of the Shapiro-Wilk test is greater than 0.05, we can say the residuals are normally distributed.

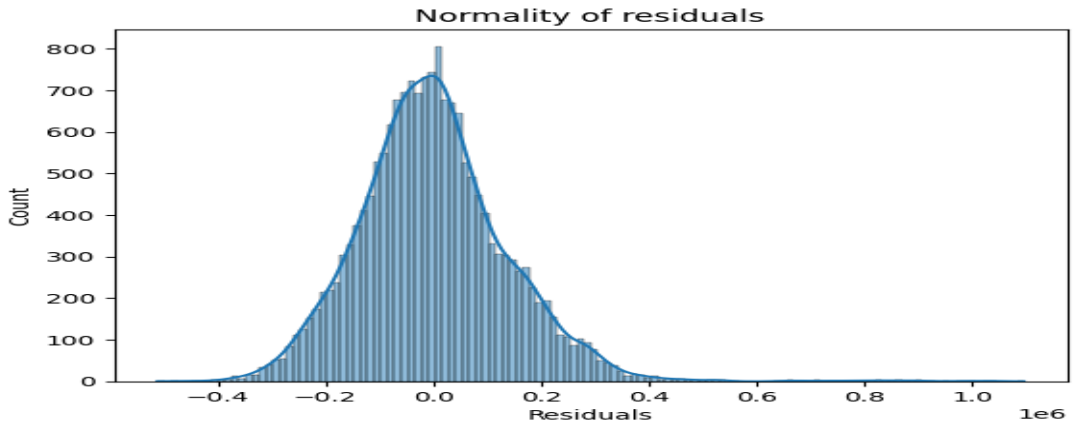

Figure 12 Normality of residuals

5.5 check Q-Q plot: -

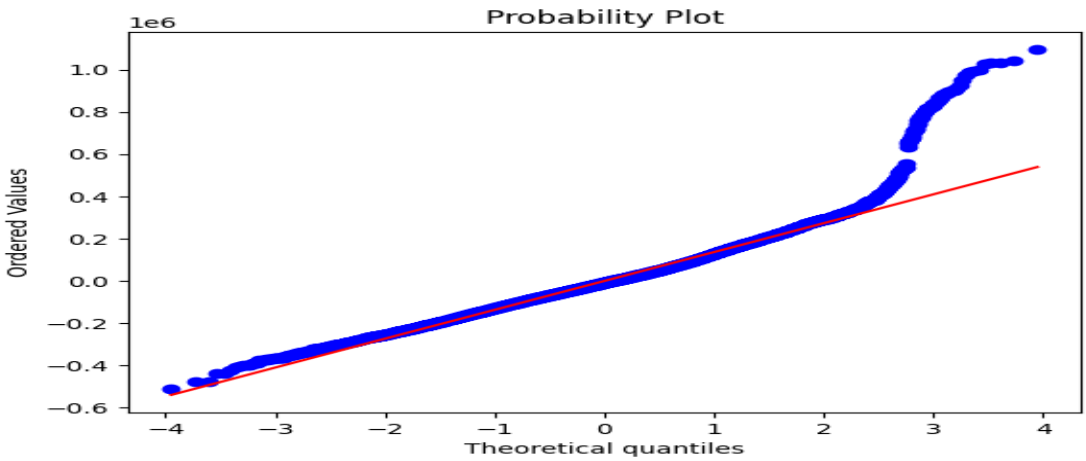

Figure 13 Probability Plot

Shapiro-Wilks test: -

The ShapiroResult gives the results of pvalue which is below 0.05 so it is good for probability.

- statistic= 0.9642057692013399
- pvalue= 1.482309045795435e-53

5.6 TEST FOR HOMOSCEDASTICITY:-

- We will test for homoscedasticity by using the goldfeldquandt test.

- If we get a p-value greater than 0.05, we can say that the residuals are homoscedastic. Otherwise, they are heteroscedastic.
- Here the p-value is >0.05 (i.e 0.808) so it is under the Test for Homoscedasticity.

```
[('F statistic', np.float64(0.9812681387454052)),
 ('p-value', np.float64(0.8087344125183018))]
```

Table 24 test for homoscedasticity:

5.7 Interpretation of the model(s)

- Linear regression model is high in all the cases like R-square has 0.986 and Adj R-square is 0.985 and all the p values are below 0.05.
- Comparing to other models it has certain performance for the probability and estimation. Accuracy also good.
- It split the data into 70:30 ratio to analyse and to train the test data.
- Below the best performance with linear training set R square and low MAPE.

Below are the training and test performances: -

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	66159.155912	46439.056764	0.996725	0.996669	3.578321

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	66400.151442	46683.134184	0.996783	0.996653	3.556775

6. Model Tuning: -

The method tuning is used to model performance improvement.

6.1 Hyperparameter Tuning - Decision Tree: -

This is Decision tree tuning model for prediction about the performance and further analysis.

```
DecisionTreeRegressor
DecisionTreeRegressor(max_depth=3, max_leaf_nodes=5, min_samples_leaf=10,
                      random_state=1)
```

Table 25 decision tree tuning

Model performance for train data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.237673	0.190423	0.359928	0.316679	134.84956

Table 26 train set

Model performance for test data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.384593	0.334702	-0.934415	-1.625278	84.687576

Table 27 test set

6.2 Hyperparameter Tuning - Gradient Boosting Regressor: -

This model has limited features as max -1, random_state is 1 and subsample also 1.

```
GradientBoostingRegressor
GradientBoostingRegressor(max_features=1, random_state=1, subsample=1)
```

Table 28 Gradient Boosting Regressor model

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.063728	0.051762	0.953981	0.950872	34.456313

Table 29 train set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.364986	0.328372	-0.742207	-1.364424	87.552859

Table 30 test set

6.3 Hyperparameter Tuning - Random Forest: -

```
RandomForestRegressor
RandomForestRegressor(max_depth=6, max_features=None, n_estimators=80,
                      random_state=1)
```

Table 31 Random Forest model

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.123372	0.099771	0.827534	0.815881	77.630671

Table 32 train set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.382785	0.339278	-0.916275	-1.600659	97.253916

Table 33 test set

6.4 Hyperparameter Tuning - XGB Regressor: -

Tuning the XGB Regressor model to improve the XGB Regressor model performance. However, This is a better model. It has least MAPE (71.95) on the test set.

```
XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=0.8, colsample_bynode=None, colsample_bytree=1,
             device=None, early_stopping_rounds=None, enable_categorical=False,
             eval_metric=None, feature_types=None, gamma=1, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=None, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None, n_estimators=100,
             n_jobs=None, num_parallel_tree=None, random_state=1, ...)
```

Table 34 XGB Regressor Hyperparameter Tuning

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.284719	0.246517	0.081448	0.019383	180.574943

Table 35 XGB Tuning Performance on Training set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.305797	0.265244	-0.222961	-0.659733	71.956117

Table 36 XGB Tuning Performance on Test set

6.5 Hyperparameter Tuning - Bagging Regressor: -

```
BaggingRegressor
BaggingRegressor(max_features=0.8, max_samples=0.9, n_estimators=90,
                 random_state=1)
```

Table 37 Bagging R model

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.120439	0.099296	0.835636	0.82453	78.651785

Table 38 Train set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.365439	0.326484	-0.746532	-1.370293	93.540917

Table 39 test set

6.6 Hyperparameter Tuning - AdaBoost Regressor

```
AdaBoostRegressor
AdaBoostRegressor(learning_rate=0.01, n_estimators=np.int64(50), random_state=1)
```

Table 40 adaboost model

Checking model performance on training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.186444	0.145386	0.606116	0.579503	110.686433

Table 41 train set

Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.186444	0.145386	0.606116	0.579503	110.686433

Table 42 test set

However, we have applied these tunings but not better results which causes to less priority.

7.0 Stacking Model: -

Now, let's build a stacking model with the tuned models - decision tree, random forest, and gradient boosting, then use XGBoost to get the final prediction.

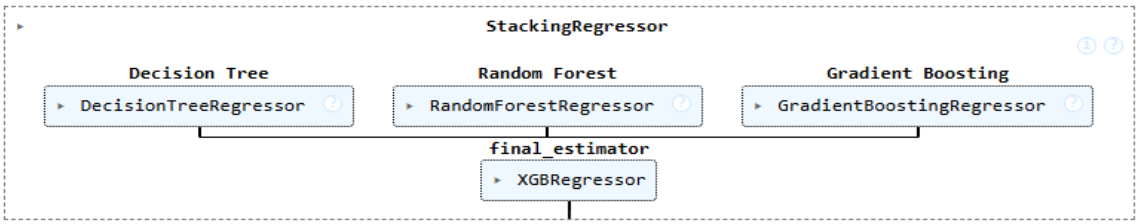

Figure 14 Stacking Model

7.1 Checking model performance on training set | Checking model performance on test set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.2809	0.211574	0.105925	0.045514	217.660347

Table 43 stacking train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.459891	0.370341	-1.766031	-2.753899	118.64715

Table 44 stacking test set

8. Model validation:-

The Models were validated not only with accuracy but also with tuning, stacking regressor and compared predicted values with actual values.

- Both training and testing metrics helps to detect overfitting or underfitting.
- The training RMSE is low but testing RMSE is high so model may be overfit.
- The XGBoost model shows strong generalization (low RMSE and MAE on test set, high R-square).
- According to the Scatter plot of Observed vs. Predicted (for XGBoost) all the points align tightly with the diagonal which is called predictions closely match real values. It's a visual sign of a highly accurate and well-calibrated model.

8.1 Training performance comparison

Training performance comparison:

	Decision Tree	Decision Tree Tuned	Bagging Regressor	Bagging Regressor Tuned	Random Forest Estimator	Random Forest Tuned	Adaboost Regressor	Adaboost Tuned	Gradient Boost Estimator	Gradient Boost Tuned	XGB	XGB Tuned	Stacking Regressor
RMSE	0.0	13959.606109	7834.930871	22714.313486	6654.919681	17246.978858	172635.936789	161257.568734	35272.267702	17635.065362	12680.401255	9973.419073	19959.181546
MAE	0.0	5792.759355	2718.694410	15174.645725	2007.206498	7105.470800	137687.553538	128397.183092	18960.784478	11193.892862	8055.084961	6534.388184	8542.606445
R-squared	1.0	0.999855	0.999954	0.999616	0.999967	0.999779	0.977845	0.980669	0.999075	0.999769	0.999880	0.999926	0.999704
Adj. R-squared	1.0	0.999853	0.999954	0.999611	0.999967	0.999776	0.977518	0.980383	0.999061	0.999765	0.999879	0.999925	0.999700
MAPE	0.0	0.495083	0.244128	0.825048	0.203754	0.633753	9.837580	8.813870	1.414626	0.828436	0.612859	0.495473	0.689119

Table 45 Comparison of training set

8.2 Testing performance comparison

Testing performance comparison:

	Decision Tree	Decision Tree Tuned	Bagging Regressor	Bagging Regressor Tuned	Random Forest Estimator	Random Forest Tuned	Adaboost Regressor	Adaboost Tuned	Gradient Boost Estimator	Gradient Boost Tuned	XGB	XGB Tuned	Stacking Regressor
RMSE	20881.902055	20344.243871	18016.597314	50099.509343	16804.975652	20512.176656	172707.519049	160256.470161	35112.108172	25785.034630	23492.712061	23280.144329	22821.177533
MAE	7478.425200	8991.549836	6248.454920	34896.133952	5343.361620	8832.173980	137922.916014	127710.915209	19032.184136	13550.837252	11638.579102	11116.125977	10074.648438
R-squared	0.999678	0.999694	0.999760	0.998144	0.999791	0.999689	0.977946	0.981011	0.999088	0.999508	0.999592	0.999599	0.999615
Adj. R-squared	0.999658	0.999675	0.999745	0.998029	0.999778	0.999670	0.976583	0.979837	0.999032	0.999478	0.999567	0.999575	0.999591
MAPE	0.657026	0.718240	0.548788	1.855715	0.501264	0.700579	9.557833	8.428898	1.324541	1.012132	0.916836	0.895143	0.781118

Table 46 Comparison of testing set

- The scores on the test data are better for tuned decision tree, bagging, random forest, gradient boost, xgboost and stacking regressor models than the other models.
- However, the model is generalizing the best in the xgboosting estimator model.
- So, we are choosing the xgboosting estimator model as the best model.

8.3 Plot observed and predicted values of the test data for the best model i.e. XGBoost Regressor model: -

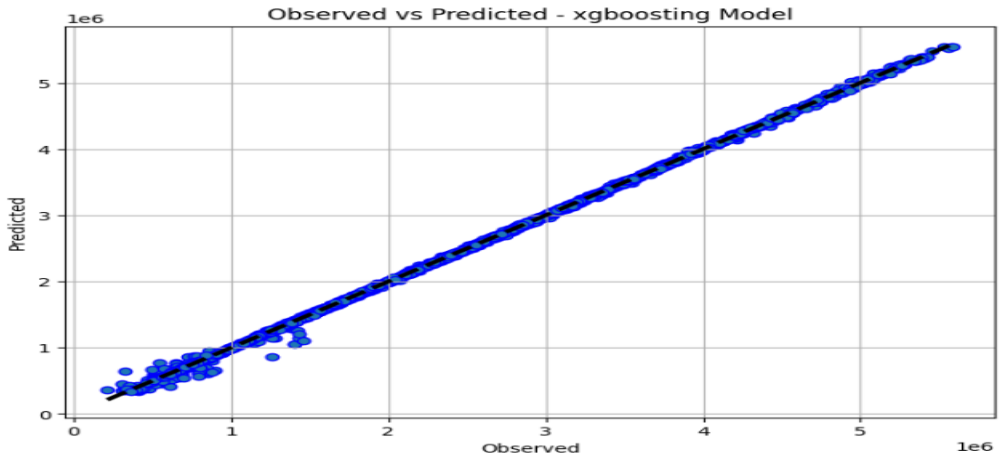

Figure 15 Observed vs Predicted

- We can see that points are dense on the line where predicted is equal to the observed.
- This implies that most of the predicted values are close to the true values with some exceptions as seen in the plot.

9. Final interpretation / recommendation: -

9.1 Interpretations: -

- Best overall performance on both training and testing data.
- Lowest RMSE (23,280) and MAPE (0.89%) on test data.
- Highest R^2 (0.9996) indicating exceptional predictive power.
- Finally, Consistent and reliable across unseen data which is a low risk of overfitting.
- Candidates rated as Key performers are 4191 count which is a good sign.
- Experience with above 3.5 years working with less salaries comparatively below

9.2 Recommendations: -

- The Tuned XGBoost model can deploy in the production environment.
- Regular performance monitoring setup to track prediction accuracy and to detect any data drift overtime.
- As a secondary backup model use Stacking Regressor since it has performed very good and will be helpful for redundancy.
- Overall, implementation of XGBoost model ensure highly accurate predictions with least error, better to management decision-making, resource planning, financial forecasting and preparation of budget.
- Focus on Key performers whose main roles and responsibilities are very close to the requirement. Therefore, it will increase the production accurately and able to assist on the other areas as well.
- There are experienced candidates over 3.5 years working for low salaries so hiring process will filter the skilled one's and will offer the salaries at minimum range which is better than their current CTC.